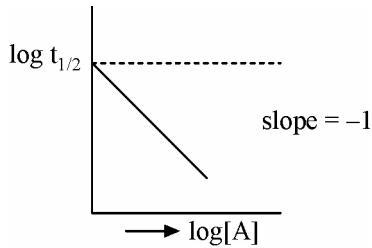


illuminati - 2019

Advanced Chemistry Assignment - 4B | Physical Chemistry Class-XIIth

CHEMICAL KINETICS & RADIOACTIVITY

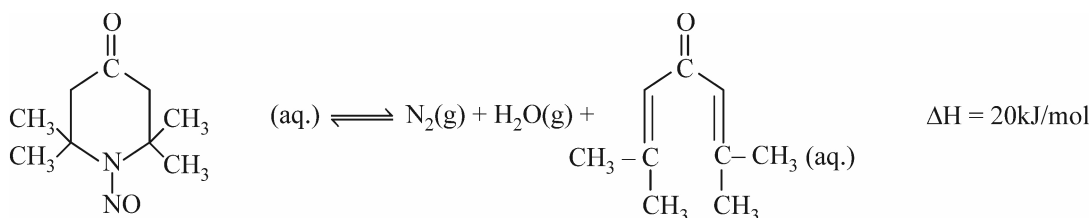
SINGLE CHOICE CORRECT QUESTION

- If the reaction $A + B \longrightarrow P$ is exothermic to the extent of 30 kcal/mole and activation energy for the forward reaction is 294 kJ/mole. Then activation energy for the backward reaction in kcal/mole is :
 (A) 324 (B) 264 (C) 100 (D) 40
 - A solid particle having spherical shape is assumed to react only on its surface. However the amount of reactant contained in it is determined by its volume. Consider the following rate law for a solid particle shrinking due to reaction $-\frac{dV}{dt} = kS$ [where S = Surface area]
 The order of the above reaction will be :
 (A) 0 (B) 1 (C) $\frac{2}{3}$ (D) $\frac{3}{2}$
 - A sample of rock from South America contains equal moles of U^{238} and Pb^{206} . The age of rock is :
 ($t_{1/2}$ for $U^{238} = 4.5 \times 10^9$ year).
 (A) 4.5×10^9 year (B) 9×10^9 year (C) 13.5×10^9 year (D) None of these
 - The graph of $\log t_{1/2}$ v/s $\log[A]$ is given below for the reaction :
 $A \longrightarrow \text{product}$
 The rate law for the reaction is :
 (A) $-\frac{d[A]}{dt} = k$ (B) $-\frac{d[A]}{dt} = k[A]$
 (C) $-\frac{d[A]}{dt} = k[A]^2$ (D) $-\frac{d[A]}{dt} = k[A]^3$
- 

The graph shows a straight line with a negative slope. The y-axis is labeled $\log t_{1/2}$ and the x-axis is labeled $\log[A]$. A dashed horizontal line extends from the y-axis to the line, and a label 'slope = -1' points to the line.
- Consider the following reaction occurring in gas phase : $CHCl_3(g) + Cl_2(g) \longrightarrow HCl(g) + CCl_4(g)$
 The rate law for the reaction is experimentally found as : $\text{Rate} = k[CHCl_3][Cl_2]^{1/2}$
Step - 1 : $Cl_2(g) \rightleftharpoons 2Cl(g)$ **Step - 2 :** $Cl(g) + CHCl_3(g) \longrightarrow HCl(g) + CCl_3(g)$
Step - 3 : $Cl(g) + CCl_3(g) \longrightarrow CCl_4(g)$
 Based on this information, what conclusion can be drawn about this proposed mechanism :
 (A) Step 1 is RDS (B) ~~Step 2 is RDS~~
 (C) Step 3 is RDS (D) RDS cannot be identified
 - A mixture of 2 gases 'R' and 'X' is kept in container at 300 K. X is inert and 'R' decomposes according to the following first order reaction: $R(g) \longrightarrow 2A(g) + 3B(g) + 4C(s)$
 Initial pressure of the reaction in a experiment is recorded as 210 torr. After 10 min, pressure is 330 torr and after a very long times 430 torr. The half life of 'R' is : [Given : Vapour Pressure of C = 20 torr under given condition]
 (A) 15 min (B) 10 min (C) 12 min (D) Data insufficient
 - A radioactive substance A_ZX undergoes α decay to produce ${}^{A-4}_{Z-2}Y$. If 1 gm atom A_ZX produced 67.2 L of He gas at 1 atm and 273 K conditions in 34.65 hour then the disintegration constant of A_ZX in (hour^{-1}) is :
 (A) 2×10^{-2} (B) 4×10^{-2} (C) 8×10^{-2} (D) 3.6×10^{-3}

Paragraph for Questions 8 - 10

Consider the interconversion of nitrosotriacetamine into nitrogen, phorone and water

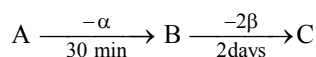


The reaction is 1st order in both directions, with an equilibrium constant of 10^4 , the activation energy for the forward reaction is 57.45 kJ/mol. Assuming Arrhenius pre-exponential factor of forward reaction 10^{12} sec^{-1} .

8. What is expected forward rate constant at 300 K, if we initiate this reaction starting with only reactant ?
 (A) 10^2 (B) 10^6 (C) 10^8 (D) 10^4
9. If the change in entropy of the reaction is $0.07 \text{ kJ K}^{-1} \text{ mol}^{-1}$ at 1 atm pressure. The temperature upto which forward reaction would not be spontaneous is :
 (A) $T < 285.7 \text{ K}$ (B) $T > 200 \text{ K}$ (C) $T > 340.2 \text{ K}$ (D) $T < 200 \text{ K}$
10. The rate constant for backward reaction at 300 K is :
 (A) 10^{-2} (B) 10^2 (C) 10^4 (D) None of these

MULTIPLE CHOICE QUESTIONS

11. A radioactive element "A" decays by the following sequence and half life as shown below :

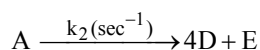
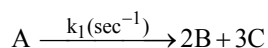


Which is/are correct ?

- (A) after 2 hr A will be less than 10% of the initial amount.
 (B) Amount of B present at any time is less than 90% of initial amount of A.
 (C) Atomic numbers of A and C are same.
 (D) The mass number of B is greater than A
12. For a reaction mechanism $A_2 \xrightleftharpoons[k_2]{k_1} 2A \text{ (fast)} ; A + B \xrightarrow{k_3} \text{Product (slow)}$

Which of the following relations are correct ?

- (A) $\text{Rate} = k_3 \left(\frac{k_1}{k_2} \right)^{1/2} [A_2]^{1/2} [B]$ (B) $\text{Rate} = k_3 \sqrt{\frac{k_1}{k_2} \times k_{eq}} [A_2]^{1/2} [B]$
 (C) $\frac{d[B]}{dt} = k_3 k_{eq}^{1/2} [A_2]^{1/2} [B]$ (D) $\frac{d[A]}{dt} = k_3 \left(\frac{k_1}{k_2} \right)^{1/2} [A_2]^{1/2} [B]$
13. $A \xrightarrow{k_A} \text{product}, B \xrightarrow{k_B} \text{product}$
 Two reactions with half life periods T_A and T_B (in min) and $k_A(\text{min}^{-1})$ and $k_B(\text{mol L}^{-1} \text{ min}^{-1})$. If half life periods are equal at the start of the reaction, the reaction rate ratio at the start of reaction would be [assuming initial concentrations of A and B are same] [Given : $\ln 2 = 0.693$]
 (A) $2k_A T_A$ (B) 1.386 (C) $\frac{k_A T_A}{2}$ (D) 0.693
14. For given parallel reaction

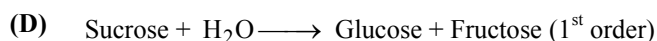
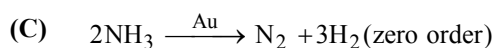
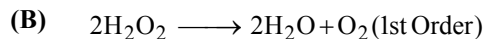
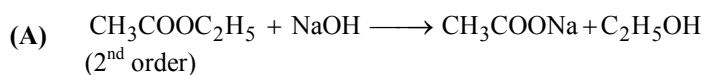


Which of the following is/are correct for % yield of mole in product mixture?

- (A) $\%B = \frac{40k_1}{k_1 + k_2}$ (B) $\%D = \frac{80k_2}{k_1 + k_2}$ (C) $\%C = \frac{60k_1}{k_1 + k_2}$ (D) $\%E = \frac{10k_2}{k_1 + k_2}$

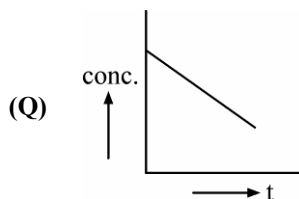
15. Match the following

Column I



Column II

(P) $k = A e^{-\frac{E_a}{RT}}$



(R) $\text{Rate} = r_0 e^{-kt}$

(S) $t_{1/2}$ is constant w.r.t. to conc. of Reactant

16. Match the following

Column I

(A) Rate constant has the same unit as the rate of reaction.

(B) Reactions having apparent molecularity more than three

(C) Reactions having molecularity two but order of reaction is one

(D) For a reaction, $\text{A} \rightarrow \text{B}$ the rate of reaction doubles as the concentration of A is doubled

Column II

(P) First order reaction

(Q) Zero order reaction

(R) Complex reaction

(S) Pseudo unimolecular reaction.

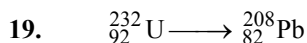
SUBJECTIVE QUESTIONS

17. For reaction $\text{A} \longrightarrow \text{B}$, following data observed.

Time	0	5	15
$\left[\frac{1}{\text{A}}\right]$	1	2	4

Determine the order of reaction

18. For a chemical reaction, $\text{A} \longrightarrow \text{Products}$, rate increases 4 times when catalyst is used. The difference in Activation energy (in kJ/mol) of without & with catalysts at 300K is X. Find 2X?



Calculate difference between number of α particles and β particle emitted in the process.

20. A freshly prepared radioactive sample of half life 60 min emits radiation of intensity which is 64 times the permissible safe level. Calculate the minimum time (in hrs) after which it would be possible to work safely with the source.

21. Under same condition initial conc. of 2.772 M of reactant A becomes one-fourth in 100 sec and in 50 sec through first order and zero order kinetics respectively. Calculate the value of Z. (Where $Z = \frac{k_0}{k_1}$)

k_0 = rate constant for zero order kinetics.

k_1 = rate constant for 1st order kinetics.

22. For first order reaction the time required for 93.75% of reaction completion is x times to the time required for half of the reaction. The value of (x) is _____.

ELECTROCHEMISTRY

SINGLE CHOICE QUESTION

23. The reduction potential of water at 25°C , in a buffer of $\text{pH} = 10$ is :
(Assuming partial pressure of H_2 gas at $25^{\circ}\text{C} = 1 \text{ atm}$)
(A) $+0.5910 \text{ V}$ (B) -0.5910 V (C) -0.8274 V (D) $+0.8274 \text{ V}$
24. **Statement : I** Electrochemical cell corresponds to reaction $\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightleftharpoons \text{H}_2\text{O}(\ell)$ is a concentration cell.
Statement : II In above reaction H_2 gas can be used at same pressure and same pH of solutions at both the electrodes.
(A) Statement-I is True, Statement-II is True and Statement-II is a correct explanation for Statement-I.
(B) Statement-I is True, Statement-II is True and Statement-II is NOT a correct explanation for Statement-I.
(C) Statement-I is True, Statement-II is False.
(D) Statement-I is False, Statement-II is True.
25. pK_a of a weak monobasic acid HA is 10. The limiting ionic conductances of H^+ and A^- ions are $350 \text{ Scm}^2 \text{ mol}^{-1}$ and $40 \text{ Scm}^2 \text{ mol}^{-1}$ respectively. The molar conductance of 10^{-2} M aq. solution of HA will be :
(A) $3.9 \times 10^{-2} \text{ Scm}^2 \text{ mol}^{-1}$ (B) $3.9 \times 10^{-4} \text{ Scm}^2 \text{ mol}^{-1}$
(C) $3.9 \times 10^{-1} \text{ Scm}^2 \text{ mol}^{-1}$ (D) $3.9 \times 10^{-3} \text{ Scm}^2 \text{ mol}^{-1}$
26. For the following cell at 298 K , $\text{Zn} \left| \text{Zn}^{2+}(\text{aq}) \right|_{0.01 \text{ M}} \text{HCl}(\text{aq}) \left| \text{H}_2(\text{g}) \right|_{1 \text{ atm}} \text{Pt}$, with $E_{\text{cell}} = 0.701 \text{ V}$ and $E_{\text{Zn}^{2+}|\text{Zn}}^0 = -0.76 \text{ V}$. Which of the following amounts of NaOH will just make the pH of cathodic compartment to be equal to 7.0.
(A) 0.4 g (B) 4 g (C) 10 g (D) 2 g
27. The conductivity of 0.01 M HA is $3.8 \times 10^{-5} \Omega^{-1} \text{ cm}^{-1}$ and conductivity of solution formed by mixing 100 cm^3 of 0.01 M HA and 1 cm^3 of 1 M NaOH is $80.0 \times 10^{-5} \Omega^{-1} \text{ cm}^{-1}$. Calculate $\lambda_{\text{A}^-}^{\infty}$, if $\lambda_{\text{Na}^+}^{\infty} = 50 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$
(A) $76.2 \times 10^{-5} \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$ (B) $76.2 \times 10^{-5} \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$
(C) $30.0 \times 10^{-5} \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$ (D) $30.0 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$
28. Consider the four different cells
 $\text{Ag} | \text{Ag}^+(\text{Ag}_2\text{CrO}_4, \text{ saturated sol.}) || \text{Ag}^+(0.1 \text{ M}) | \text{Ag}$
 $E_{\text{cell}} = x \text{ V}$ and $K_{\text{sp}, \text{Ag}_2\text{CrO}_4} = 'p'$
 $\text{Ag} | \text{Ag}^+(\text{AgCl saturated sol.}) || \text{Ag}^+(0.1 \text{ M}) | \text{Ag}$
 $E_{\text{cell}} = x \text{ V}$ and $K_{\text{sp}, \text{AgCl}} = 'q'$
 $\text{Ag} | \text{Ag}^+(\text{Ag}_3\text{PO}_4 \text{ saturated sol.}) || \text{Ag}^+(0.1 \text{ M}) | \text{Ag}$
 $E_{\text{cell}} = x \text{ V}$ and $K_{\text{sp}, \text{Ag}_3\text{PO}_4} = 'r'$
 $\text{Pt} | \text{H}_2(\text{g}), 1 \text{ bar} | \text{H}^+(1 \text{ M}) || (\text{Ag}_2\text{SO}_4 \text{ saturated sol.}) \text{Ag}^+ | \text{Ag}$
 $E_{\text{cell}} = x \text{ V}$ and $K_{\text{sp}, \text{Ag}_2\text{SO}_4} = 's'$
The correct option for solubility product for all the four cases is (by considering all cells are in equilibrium stage).
(A) $s > q > p > r$ (B) $s > p > q > r$ (C) $r > s > q > p$ (D) $r > q > p > s$

29. Choose the correct statement(s) from the following :
- (i) Van't Hoff factor for 0.001 N CH_3COOH ($\Lambda^0 = 350 \text{ S cm}^2 \text{ mol}^{-1}$ & $k = 4 \times 10^{-5} \text{ S cm}^{-1}$) is 39/35
- (ii) If Λ^0 of KCl , LiCl and KNO_3 are 150, 115 & $145 \text{ S cm}^2 \text{ mol}^{-1}$ then that of LiNO_3 will be $110 \text{ S cm}^2 \text{ mol}^{-1}$.
- (iii) In general with increase in concentration, specific conductance increases and reaches a maximum & then decreases with further increase in concentration.
- (A) (i), (ii) and (iii) (B) (i) and (ii) (C) Only (i) (D) Only (ii)

Paragraph for Questions 30 - 31

According to Kohlrausch law at infinite dilution each ion makes a definite contribution towards molar conductance of electrolyte irrespective of the nature of the other ion with which it is associated and $\lambda_m^0 = \lambda_+^0 + \lambda_-^0$ although at infinite dilution, all electrolytes dissociate completely but their conductance differ from each because of the difference in their mobilities, defined as the distance travelled by an ion per sec under a potential gradient of 1V/m.

30. The degree of dissociation of pure water at 55°C is 2.8×10^{-8} . Then the specific conductance of water (in mho m^{-1}) at this temperature will be : [Given $\lambda_{\text{H}^+}^0 = 349.83 \text{ mho cm}^2 \text{ mol}^{-1}$; $\lambda_{\text{OH}^-}^0 = 198.5 \text{ mho cm}^2 \text{ mol}^{-1}$]
- (A) 7.4×10^{-8} (B) 3.2×10^{-4} (C) 7.2×10^{-6} (D) 8.5×10^{-5}
31. A dilute solution of NaCl is placed between two Pt electrodes which are 8.5 cm apart. A potential difference of 7.5 volt is applied across the electrodes, what will be the distance travelled by Na^+ in 1 Hr ?
- [Given : $\lambda_{\text{Na}^+}^0 = 70 \text{ S cm}^2 \text{ mol}^{-1}$]
- (A) 2.3 cm (B) 4.8 cm (C) 3.5 cm (D) 7.8 cm

Paragraph for Questions 32 - 33

People having amalgam filling in their teeth face various problems. If Al comes in contact with an amalgam filling it causes a sharp tooth pain. The amalgam filling, an alloy of Ag, Sn and Hg acts as cathode of tiny galvanic cell. Al behaves as anode and saliva acts as electrolyte. When the Al and amalgam come in contact, an electric current passes from Al to the filling, which is sensed by the nerves in the tooth. Al is oxidized at anode and O_2 gas is reduced to H_2O at cathode

[Given : $E_{\text{Al}^{3+}|\text{Al}}^\circ = -1.66 \text{ V}$; $E_{\text{O}_2, \text{H}^+|\text{H}_2\text{O}}^\circ = 1.23 \text{ V}$]

32. Net cell reaction taking place when amalgam is in contact with Al foil.
- (A) $2\text{H}_2 + \text{O}_2 \longrightarrow 2\text{H}_2\text{O}$
- (B) $4\text{Al} + 3\text{O}_2 + 12\text{H}^+ \longrightarrow 4\text{Al}^{3+} + 6\text{H}_2\text{O}$
- (C) $4\text{Al} + 3\text{O}_2 + 6\text{H}_2\text{O} + 4\text{OH}^- \longrightarrow 4[\text{Al}(\text{OH})_4]^-$
- (D) $4\text{Al} + 3\text{O}_2 \longrightarrow 2\text{Al}_2\text{O}_3$
33. E° experienced by the person with dental filling of Ag, Sn and Hg amalgam is :
- (A) -2.89 V (B) -0.93 V (C) $+2.89 \text{ V}$ (D) 0.43 V

MULTIPLE CHOICE QUESTIONS

34. The standard electrode potentials E° of the following half cell reactions are :
- (i) $\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \longrightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$ $E^\circ = 1.51 \text{ V}$
- (ii) $\text{Sn}^{2+} \longrightarrow \text{Sn}^{4+} + 2\text{e}^-$ $E^\circ = -0.15 \text{ V}$
- (iii) $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \longrightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$ $E^\circ = 1.33 \text{ V}$
- (iv) $\text{Ce}^{3+} \longrightarrow \text{Ce}^{4+} + \text{e}^-$ $E^\circ = -1.61 \text{ V}$

The oxidizing power of the various species are related as :

- (A) $\text{Cr}_2\text{O}_7^{2-} > \text{MnO}_4^-$ (B) $\text{Ce}^{4+} > \text{Sn}^{4+}$ (C) $\text{Ce}^{4+} > \text{MnO}_4^-$ (D) $\text{MnO}_4^- > \text{Sn}^{4+}$

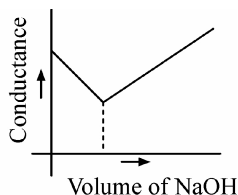
35. Given : $E^\circ_{\text{Ag}^+|\text{Ag}} = 0.80 \text{ V}$; $E^\circ_{\text{Mg}^{2+}|\text{Mg}} = -2.73 \text{ V}$; $E^\circ_{\text{Cu}^{2+}|\text{Cu}} = 0.34 \text{ V}$; $E^\circ_{\text{Hg}^{2+}|\text{Hg}} = 0.79 \text{ V}$;

Select the incorrect statements

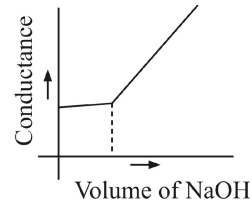
- (A) AgNO_3 can be stored in copper vessel (B) $\text{Cu(NO}_3)_2$ can be stored in magnesium vessel
(C) CuCl_2 can be stored in silver vessel (D) HgCl_2 can be stored in copper vessel

36. Which is/are correct match about conductometric titration ?

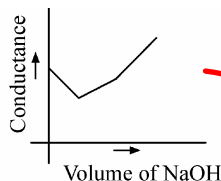
(A) HCl vs NaOH



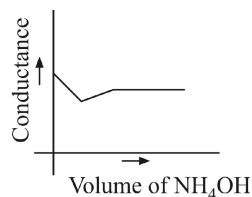
(B) CH_3COOH vs NaOH



(C) $(\text{HCl} + \text{CH}_3\text{COOH})$ vs NaOH



(D) $(\text{CH}_3\text{COOH} + \text{HCl})$ vs NH_4OH



COLUMN MATCHING TYPE

37. Match the column

Column I

(A) $\text{Zn} | \text{Zn}^{2+} (1\text{M}) || \text{Zn}^{2+} (2\text{M}) | \text{Zn}$

(B) $\text{Pt} | \text{H}_2(\text{g}) | \text{HCl} (1\text{M}) || \text{H}_2\text{SO}_4 (1\text{M}) | \text{H}_2(\text{g}) | \text{Pt}$
1 atm 1 atm

(C) $\text{Cu} | \text{Cu}^{2+} (0.01\text{M}) || \text{Ag}^+ (0.1\text{M}) | \text{Ag}$

(D) $\text{Ag} | \text{AgCl} | \text{KCl} (0.1\text{M}) || \text{Ag}^+ (0.01\text{M}) | \text{Ag}$

Column II

(P) Spontaneous cell reaction

(Q) Working cell representation

(R) Concentration of cation in cathodic compartment increases upto equilibrium conc.

(S) Concentration cell.

Given : $E^\circ_{\text{Cu}^{2+}|\text{Cu}} = 0.34 \text{ V}$, $E^\circ_{\text{Ag}^+|\text{Ag}} = 0.8 \text{ V}$, $K_{\text{sp}}(\text{AgCl}) = 10^{-10}$

SUBJECTIVE QUESTIONS

38. 100 ml of 0.005 M CuSO_4 solution is electrolysed by passing a current of 0.02 amp with a current efficiency of 80% for 200 min. Calculate pH of solution after completion of electrolysis.

39. $\text{Pt}, \text{H}_2(\text{g}) | 10^{-4} \text{M NaOH} || \text{CH}_3\text{COOH} (1\text{M}) + \text{CH}_3\text{COONa} (1\text{M}) | \text{H}_2(\text{g}), \text{Pt}$
1 atm 1 atm

Calculate the value of $10 \times E_{\text{cell}}$ [Given : $\frac{2.303 RT}{F} = 0.06$, $\text{pK}_a(\text{CH}_3\text{COOH}) = 5$]

LIQUID SOLUTION, SOLID STATE & SURFACE CHEMISTRY

SINGLE CHOICE QUESTION

40. 30.11×10^{22} unit cell of K_2S is dissolved in 10 L water. K_2S has antifluorite structure. The freezing point of the solution becomes $[K_f(\text{water}) = 1.86 \text{ K kg mol}^{-1}, T_f(\text{H}_2\text{O}) = 273.19]$
 (A) 272.78 K (B) 272.034 K (C) 272.87 K (D) 272.9 K
41. Coagulation occurs when
 (A) Electrolyte added to Lyophobic colloidal solution.
 (B) Two oppositely charged colloids mixed.
 (C) Electrophoresis take place.
 (D) All of these
42. 1 molal urea solution freezes at 1.86°C and depression in freezing point of 0.01 molal aqueous CH_3COOH solution is 0.02046°C . The pH of CH_3COOH solution is:
 (Assume molality equal to molarity)
 (A) 2 (B) 3 (C) 3.2 (D) 4.2
43. The Henry's law constant for solubility of gas A in water at 25°C is 10^4 atm . The % by volume of gas A is 50% in gaseous mixture. The number of moles of A, dissolved in 100 mol of water at 10 atm pressure and 298 K is :
 (A) 5×10^{-4} (B) 10×10^{-4} (C) 5×10^{-2} (D) 10×10^{-2}
44. Which is incorrect statement?
 (A) Colourless crystal like NaCl appear to be coloured due to F-centre
 (B) In Frenkel defect, density of crystal remains constant
 (C) In Pyroelectricity electric current is produced when a crystal is subjected to mechanical stress
 (D) Above curie temperature ferromagnetism disappears in Fe.

Paragraph for Questions 45 - 47

Coagulation is the process by which dispersed phase of a colloid is made to aggregate and separate from dispersed medium. Coagulation value is defined as millimoles of electrolyte required to cause coagulation of one litre colloidal solution. Higher is coagulation power of electrolyte smaller will be the coagulation value of electrolyte.

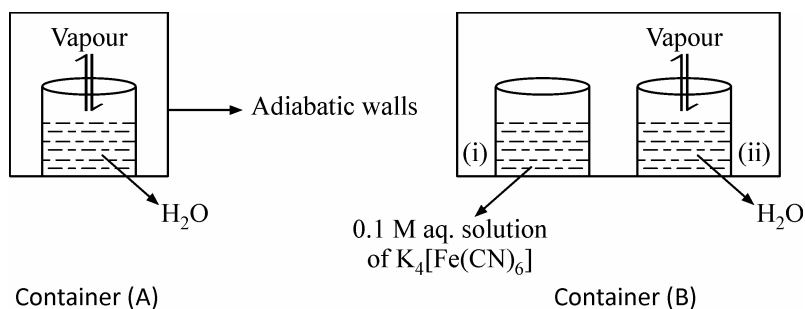
45. Which ion has highest coagulation value?
 (A) K^+ (B) Ca^{2+} (C) Al^{3+} (D) Sn^{4+}
46. For the coagulation of 200 ml of As_2S_3 solution, 10 ml of 1M NaCl solution is required. What is the coagulating value of NaCl?
 (A) 200 (B) 100 (C) 50 (D) 25
47. For the coagulation of As_2S_3 , which of the following will have maximum coagulating power?
 (A) NaCl (B) CaCl_2 (C) AlCl_3 (D) $\text{K}_4[\text{Fe}(\text{CN})_6]$

Paragraph for Questions 48 - 50

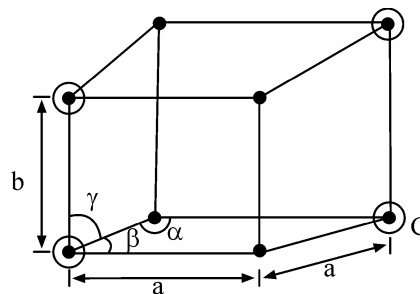
Spinel is a mineral MgAl_2O_4 having general formula $\text{MM}'_2\text{O}_4$ where divalent ions occupy the tetrahedral voids and trivalent ions in octahedral voids and lattice is made up of oxide ion present in CCP arrangement. Only $1/8$ of tetrahedral and $1/2$ of octahedral void are occupied. (Assume ideal arrangement of particles)

48. The packing fraction of MgAl_2O_4 unit cell is :
 (A) 0.768 (B) 0.786 (C) 0.74 (D) 0.68
49. The % of Mg in MgAl_2O_4 is
 (A) 16.9 % (B) 38% (C) 12.5% (D) None of these
50. If 50% octahedral voids occupied by Al^{3+} and 12.5% tetrahedral void occupied by Mg^{2+} , then formula of compound is:
 (A) MgAl_2O_4 (B) $\text{Mg}_4\text{Al}_3\text{O}_4$ (C) $\text{Mg}_2\text{Al}_2\text{O}_5$ (D) None of these

Paragraph for Questions 51 - 52



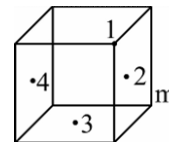
51. If we create vacuum and maintain it in container A at 25° C then :
- Water will be completely vaporized in the container A
 - There will be no vaporization in the container A
 - The equilibrium will establish in between water vapour and liq. H₂O and vapour pressure of water will remain same as it would be in presence of atmospheric pressure.
 - The equilibrium in water H₂O(g) and H₂O(l) will establish but vapour pressure of H₂O(g) will be more than as it would be in presence of atmospheric pressure.
52. Which of the following statement is correct regarding container B ?
- water will migrate from Glass (i) to Glass (ii) and no equilibrium will establish
 - water will migrate from Glass (i) to Glass (ii) and equilibrium will establish after some time.
 - water will migrate from Glass (ii) to Glass (i) and no equilibrium will establish
 - water will migrate from Glass (ii) to Glass (i) and equilibrium will establish after some time.
53. A body centered cubic lattice is made up of hollow spheres of B. Spheres of solid A are present in hollow spheres of B. Radius of A is half of radius of B. What is the ratio of total volume of spheres of B unoccupied by A in a unit cell and volume of unit cell ?
- $\frac{7\pi\sqrt{3}}{64}$
 - $\frac{7\sqrt{3}}{128}$
 - $\frac{7\pi}{24}$
 - None of these
54. Mg crystallizes in a lattice represented as $\alpha = 120^\circ$, $\beta = 60^\circ$ and $\gamma = 90^\circ$. There is one more atom in this lattice which can be located by moving $1/3^{\text{rd}}$ distance along AC and then upward $0.8165a$. The solid has density of 2.0 g/cc and $b = 1.663a$ then the approximate distance between nearest atoms will be :
- 3 Å
 - 4 Å
 - 5 Å
 - 6 Å



MULTIPLE CHOICE QUESTIONS

55. When 1 mole benzene ($P_{\text{benzene}}^\circ = 42 \text{ mmHg}$) and 2 mole toluene ($P_{\text{toluene}}^\circ = 36 \text{ mmHg}$) are mixed. Then select correct statement(s) :
- Mole fraction of C₆H₆ in vapour phase is 7/19
 - Mole fraction of C₆H₆ in liquid phase is 1/3
 - Total vapour pressure of mixture is 38 mmHg
 - Benzene and toluene form ideal solution.
56. Which is true regarding colloidal solution ?
- The boiling point of colloidal solution is smaller than the true solution under similar conditions.
 - Colloidal particles can be coagulated by addition of strong electrolyte.
 - Brownian movement is shown by colloidal solution
 - Sodium salt of fatty acids form colloids at higher concentration.

57. Which of the following statements is/are false for physisorption ?
 (A) High temperature is favourable for adsorption
 (B) It needs activation energy.
 (C) It is reversible.
 (D) Extent of adsorption increases with increase in pressure.
58. Select the correct statements :
 (A) $\text{Fe}(\text{OH})_3$ sol can be used for coagulation of As_2S_3 sol
 (B) Aqueous gold sol is negatively charged sol.
 (C) Langmuir adsorption is a single layer phenomenon.
 (D) Activity of an enzyme is pH dependent
59. 2.5 gm of a monobasic acid HA in 100 gm water lowers the freezing point by 0.2325 K. If 0.75 gm of same acid requires $15\text{ml } \frac{1}{5}\text{M NaOH}$ solution for complete neutralization. Which is/are correct statement ?
 ($K_f = 1.86\text{ K kg mol}^{-1}$; $K_B = 0.52\text{ K kg mol}^{-1}$)
 (A) Molecular weight of acid is 250
 (B) Degree of ionization of acid is 25%
 (C) Vant Hoff factor of HA is 1.25
 (D) Difference in B.P. and M.P. is 100.2975°C
60. Which is/are correct statement ?
 (A) Flocculation value is a measure of protective power
 (B) Brownian movement is related to the stability of colloids.
 (C) When AgNO_3 is added to excess of KI then negative charge colloid results.
 (D) Coagulation takes place due to electrophoresis.
61. Consider following solutions :
 (i) 1 M aqueous glucose solution
 (ii) 1 M aqueous sodium chloride solution
 (iii) 1 M aqueous ammonium phosphate solution
 (iv) 1 M benzoic acid in benzene
 Select correct statements for the above solutions.
 (A) All are isotonic solutions
 (B) III is hypertonic of I, II and IV
 (C) IV is hypotonic of I, II and III
 (D) II is hypotonic of III but hypertonic of I and IV
62. Two liquids A and B form an ideal solution. The solution has a vapor pressure of 700 torr at 80°C . It is distilled till $\frac{2}{3}$ rd of the solution is collected as condensate. The mole fraction of A (X'_A) in the condensate is 0.75 and that of the residual liquid has mole fraction of A (X_A) is 0.30. If the vapour pressure of the residual liquid at 80°C is 600 torr. Which of the following is/are true ?
 (A) The composition of the original liquid was $X_A = 0.6$
 (B) The composition of the original liquid was $X_A = 0.4$
 (C) $P_A^0 = \frac{2500}{3}\text{ torr}$
 (D) $P_B^0 = 500\text{ torr}$
63. In a F.C.C. unit cell, atoms are numbered as shown below. Identify the pair in which distance between the atoms is same. [Atoms 2, 3, 4 are present face centre]
 (A) (1 & 4) and (1 & 3)
 (B) (1 & 2) and (4 & 3)
 (C) (1 & 2) and (1 & 4)
 (D) (1 & 2) and (2 & 3)
64. The HCP and CCP structure for a given element would be expected to have :
 (A) The same co-ordination number
 (B) The same volume of unit cell
 (C) the same packing fraction
 (D) The same effective number of atoms per unit cell
65. A solution having osmotic pressure 5 atm (X) is separated by a SPM from another solution having osmotic pressure 6 atm (Y)
 (A) Solution X is hypertonic with respect to Y and solvent flows from X to Y
 (B) If additional pressure on Y is increased by 1 atm that of X by 0.5 atm, solvent flow from Y to X
 (C) Additional pressure of 1 atm when applied on solution Y stops net flow of solvent across SPM
 (D) If concentration of (non-dissociative or associative) solute in X is 0.1 M than in Y it can be 0.12 M.



66. A solution is prepared by dissolving 1.5 g of a monoacidic base into 1.5 kg of water at 372 K, which showed a depression in freezing point by 0.165°C. When 0.496 g of the same base titrated, after dissolution, required 40 ml of semimolar H₂SO₄ solution. If K_f of water is 1.86 K kg mol⁻¹, then select the correct statement(s) out of the following (assuming molarity = molality) :
- (A) The pH of the solution of weak base is 11.90.
 (B) Ionisation constant of the base is 8×10^{-2} .
 (C) The osmotic pressure of the aqueous solution of base at 372K is 2.7 atm (approx).
 (D) The base is 10% ionized in aqueous solution.

MATRIX MATCHING TYPE

67. Match the following

Column I	Column II
(A) Vapour pressure of binary solution of two miscible and volatile liquids (containing 1 mole of each) when half of the liquid solution goes into vapour phase.	(P) $\sqrt{P_A^0 P_B^0}$
(B) Vapour pressure of binary solution of two miscible and volatile liquids.	(Q) $P_A^0 + P_B^0$
(C) Vapour pressure of binary solution of two immiscible and volatile liquids.	(R) $P_B^0 + \chi_A (P_A^0 - P_B^0)$
(D) Vapour pressure of binary solution having non-volatile solute (B) and volatile solvent (A).	(S) $\frac{P_A^0 - P}{P_A^0} = \chi_B$

68. Match the following

Column I	Column II
(A) Acetone + CHCl ₃	(P) $\Delta S_{\text{mix}} > 0$
(B) C ₂ H ₅ OH + H ₂ O	(Q) $\Delta V_{\text{mix}} > 0$
(C) CH ₃ OH + CCl ₄	(R) $\Delta H_{\text{mix}} < 0$
(D) n-hexane + n-heptane	(S) $\Delta V_{\text{mix}} < 0$
	(T) Minimum boiling azeotropes

69. Match the following

Column I (Crystal system)	Column II (Types of Bravais lattice)
(A) Rhombohedral	(P) Primitive only
(B) Orthorhombic	(Q) Primitive and body centred only
(C) Tetragonal	(R) Primitive and end centred only
(D) Hexagonal	(S) Primitive, body centred, end centred and face centred

70. Match the following

Column I (Term)	Column II (Related property)
(A) Coagulation	(P) Associative colloids
(B) Dialysis	(Q) Protective colloids
(C) CMC	(R) Purification of colloids
(D) Gold number	(S) Electrolyte
	(T) Preparation of colloids

71. Match the following

Column I (Crystal structure)

Column II (Arrangement)

(A) Rock salt (NaCl)

(P) C.N : 6 : 6

(B) Fluorite (CaF₂)

(Q) C.N : 8 : 8

(C) ZnS

(R) Fraction of tetrahedral void occupied

(D) CsCl type

(S) C.N. : 8 : 4

72. Metal element "M" of radius 50 nm is crystallized in FCC format and made cubic crystal such that face of unit cell is aligned with face of cubical crystal. If total number of metal atoms of "M" at all faces of cubic crystal is 6×10^{30} , then area of one face of cubical crystal is $(A \times 10^{16})\text{m}^2$. Then the value of A is :
73. 526.3 ml of 0.5 M HCl is shaken with 0.5g of activated charcoal and filtered. The concentration of the filtrate is reduced to 0.4 M. What is the amount of adsorption (x/m) ?
74. For ABC ABC ABC closed packed structure if the distance between two successive tetrahedral void is 'X' and distance between two successive octahedral voids is 'Y' in a unit cell then $\frac{2Y\sqrt{2}}{X}$ is:
75. For body centred orthorhombic unit cell of atoms ($M = 20 \text{ amu}$) $a = \frac{1}{2} \text{ nm}$, $b = \frac{1}{5} \text{ nm}$, $c = \frac{1}{10} \text{ nm}$. Find the value of density in the form of $\frac{\text{amu}}{(\text{\AA})^3}$.
76. To 250 ml of water, x g of acetic acid is added. If 11.5% of acetic acid is dissociated, the depression in freezing point comes out 0.416. What will be the value of x?
Given : $K_f(\text{water}) = 1.86 \text{ K kg}^{-1} \text{ mol}^{-1}$ and density of water is 0.997 g ml^{-1} .

ANSWERS KEY

Advanced Chemistry Assignment-4B | Illuminati-2019

Physical Chemistry Class XIIth

1	2	3	4	5	6	7	8	9	10	11	12	13	14	
C	C	A	C	B	B	B	A	A	A	AC	ACD	AB	ABC	
15					16					17	18	19	20	21
A - P ; B - PRS ; C - P Q; D - P R S					A - Q ; B - R ; C - S; D - P					2	7	4	6	3
22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
4	B	C	A	A	D	A	A	D	A	B	C	BCD	ABD	ABCD
37					38	39	40	41	42	43	44	45	46	47
A - P Q S ; B - P Q S ; C - P Q; D – P Q S					2	3	B	D	B	C	C	A	C	C
48	49	50	51	52	53	54	55		56		57		58	
A	A	A	A	C	A	A	ABCD		ABCD		AB		ABCD	
59		60	61	62	63	64	65	66	67					
ABCD		BCD	BCD	ACD	ABD	AC	CD	ACD	A - P ; B - R ; C - Q; D - S					
68					69					70				
A - PRS ; B - P Q T ; C - P Q T; D - P					A - P ; B - S ; C - Q; D - P					A - S ; B - R ; C - P; D - Q				
71					72	73	74	75	76					
A - P ; B - S ; C - R; D - Q					1	4	4	4	3					